

Exp No: 18

GENERATION OF PWM USING TIMER INTERRUPT

OBJECTIVES

- To generate a PWM signal using the Timer 0 interrupt of the AT89C51 microcontroller.
- To understand and control the duty cycle of the PWM signal (25%, 50%, and 75%).
- To observe and verify the generated PWM waveform using the Keil Logic Analyzer.

PROGRAM

```
#include <reg51.h>

sbit PWM = P1^0;

unsigned int count = 0;

void timer0_ISR(void) interrupt 1
{
    TH0 = 0xFC;
    TL0 = 0x18;
    count++;
    if(count >= 200)
        count = 0;
    if(count < 50)
        PWM = 1;
    else
        PWM = 0;
}

void main(void)
{
    TMOD = 0x01;
    TH0 = 0xFC;
    TL0 = 0x18;
```

```

ET0 = 1;

EA = 1;

TRO = 1;

P1 = 0x00;

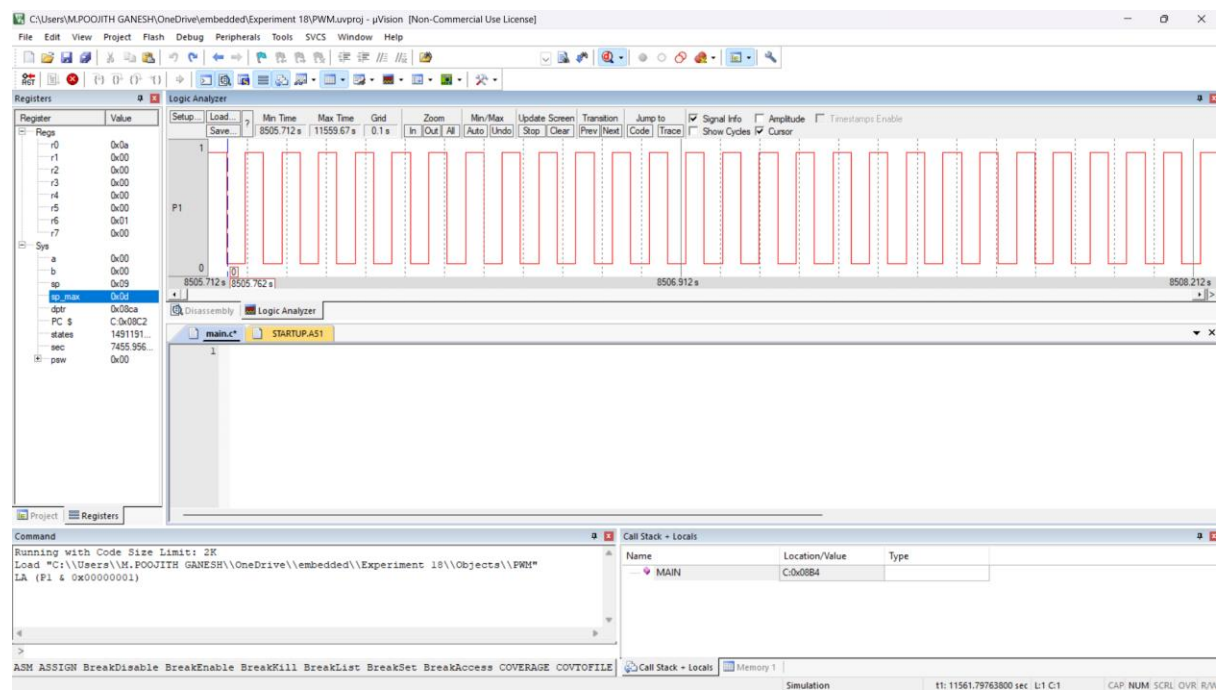
while(1);

}

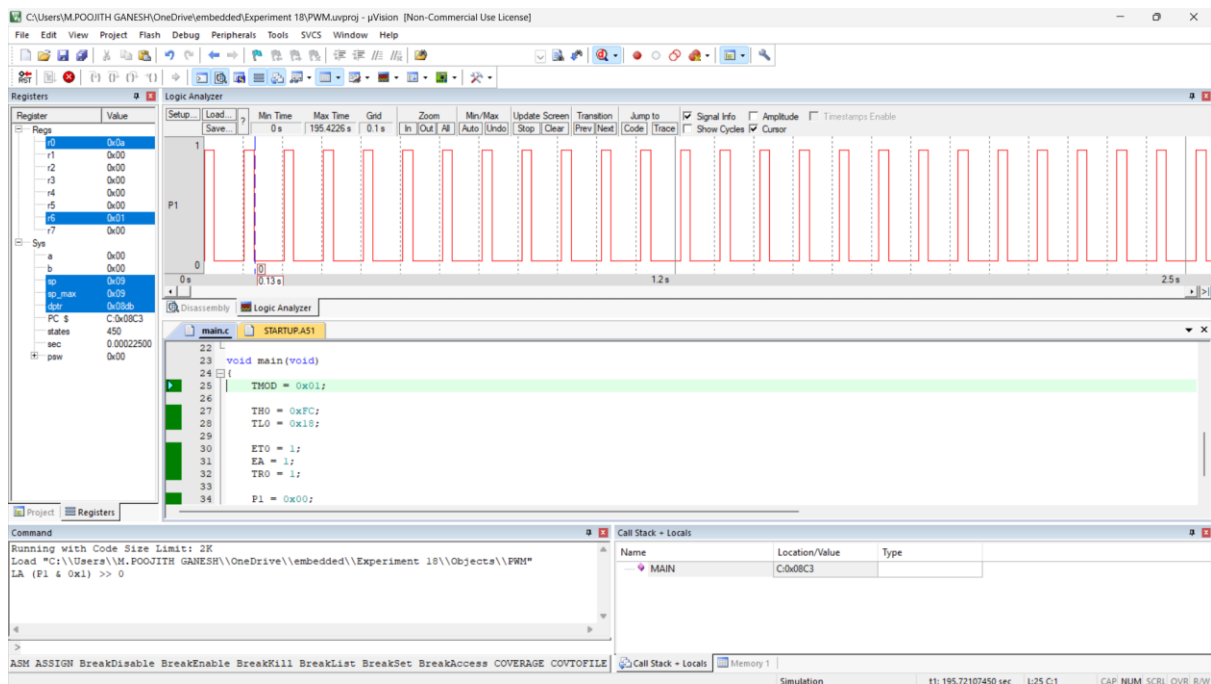
```

OUTPUT

50% Duty cycle:



25% duty cycle:



75% duty cycle:

